

RaceDay API Endpoint Plan

1. Purpose

This document defines the planned REST API endpoints for the RaceDay event management system. The endpoint plan was created during Part 1 before API implementation.

RaceDay supports two main roles:

- **Organizer** – creates and manages events, categories, routes, weather information, event images, participants, and results.
- **Participant** – browses events, manages their profile, enrolls in events, manages emergency contacts and views their performance history.

Role definitions:

- **None** – endpoint is publicly accessible.
- **Any** – endpoint requires an authenticated user of either role.
- **Organiser** – endpoint requires an authenticated Organiser.
- **Participant** – endpoint requires an authenticated Participant.

2. Endpoint Plan

HTTP		Description	Role Required	Request Body	Expected Response
#	Metho Route d				
1	POST /api/auth/register	Registers a new RaceDay user account and creates the appropriate profile based on the selected role.	None	FirstName, LastName, Email, Password, Role	201 Created – user created; 400 Bad Request – invalid data; 409 Conflict – email already exists
2	POST /api/auth/login	Authenticate s a user and returns an access token.	None	Email, Password	200 OK – JWT/token returned; 401 Unauthorized – invalid credentials

3	GET	/api/users/profile	Retrieves the profile information of the authenticated user.	Any	None	200 OK – profile returned; 401 Unauthorized
4	PUT	/api/users/profile	Updates editable information for the authenticated user's profile.	Any	FirstName, LastName, and applicable profile information	200 OK – profile updated; 400 Bad Request; 401 Unauthorized
5	GET	/api/events	Retrieves upcoming RaceDay events that participants can browse.	None	None	200 OK – list of events
6	GET	/api/events/{id}	Retrieves details for a specific event.	None	None	200 OK – event details; 404 Not Found
7	POST	/api/events	Creates a new RaceDay event.	Organiser	EventName, Description, EventDate, Location, Status	201 Created – event created; 400 Bad Request; 401 Unauthorized; 403 Forbidden
8	PUT	/api/events/{id}	Updates an event managed by the authenticated organiser.	Organiser	EventName, Description, EventDate, Location, Status	200 OK – updated event; 403 Forbidden; 404 Not Found
9	DELETE	/api/events/{id}	Deletes an event managed by the	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found

			authenticate d organiser.			
1 0	GET	/api/events/{eventId}/categories	Retrieves all categories belonging to a specific event.	None	None	200 OK – category list; 404 Not Found
1 1	POST	/api/events/{eventId}/categories	Creates a category for an event managed by the organiser.	Organise r	CategoryName, DistanceKm, EntryFee, MaxParticipants	201 Created; 400 Bad Request; 403 Forbidden; 404 Not Found
1 2	PUT	/api/categories/{id}	Updates an existing event category.	Organise r	CategoryName, DistanceKm, EntryFee, MaxParticipants	200 OK; 403 Forbidden; 404 Not Found
1 3	DELETE	/api/categories/{id}	Deletes an event category when it is not prevented by existing enrolments.	Organise r	None	204 No Content; 403 Forbidden; 404 Not Found; 409 Conflict
1 4	POST	/api/enrolments	Enrols the authenticate d participant in an event category. The participant is determined from the authentication token.	Participant	CategoryID	201 Created; 400 Bad Request; 401 Unauthorized; 404 Not Found; 409 Conflict

15	GET	/api/enrolments/my	Retrieves all event enrolments belonging to the authenticated participant.	Participant	None	200 OK – enrolment list; 401 Unauthorized
16	GET	/api/events/{eventId}/enrolments	Retrieves participants enrolled in an event for event management purposes.	Organiser	None	200 OK – participant/enrolment list; 403 Forbidden ; 404 Not Found
17	DELETE	/api/enrolments/{id}	Cancels an enrolment belonging to the authenticated participant.	Participant	None	204 No Content ; 401 Unauthorized ; 404 Not Found
18	POST	/api/results	Records a race result for an enrolment.	Organiser	EnrolmentID, FinishTime, Position, AveragePace, Status	201 Created ; 400 Bad Request ; 403 Forbidden ; 404 Not Found ; 409 Conflict
19	GET	/api/results/{id}	Retrieves a specific race result.	None	None	200 OK – result; 404 Not Found
20	GET	/api/results/my	Retrieves the authenticated participant's personal performance history.	Participant	None	200 OK – results list; 401 Unauthorized

2 1	GET	/api/events/{eventId}/results	Retrieves the results for a specific event.	None	None	200 OK – event results; 404 Not Found
2 2	GET	/api/events/{eventId}/routes	Retrieves route information for an event.	None	None	200 OK – route list; 404 Not Found
2 3	POST	/api/events/{eventId}/routes	Adds route information to an event managed by the organiser.	Organiser	RouteName, DistanceKm, StartLocation, FinishLocation, RouteUrl, ElevationGain	201 Created; 400 Bad Request; 403 Forbidden; 404 Not Found
2 4	PUT	/api/routes/{id}	Updates an existing route.	Organiser	RouteName, DistanceKm, StartLocation, FinishLocation, RouteUrl, ElevationGain	200 OK; 403 Forbidden; 404 Not Found
2 5	DELETE	/api/routes/{id}	Removes an event route.	Organiser	None	204 No Content; 403 Forbidden; 404 Not Found
2 6	GET	/api/events/{eventId}/weather	Retrieves the latest available weather information for an event.	None	None	200 OK – weather information; 404 Not Found
2 7	POST	/api/events/{eventId}/weather	Adds a weather record associated with an event.	Organiser	RecordedAt, Temperature, Condition, WindSpeed, Humidity	201 Created; 400 Bad Request; 403 Forbidden; 404 Not Found
2 8	POST	/api/payments	Records a payment attempt for a	Participant	EnrolmentID, Amount	201 Created – payment recorded; 400 Bad Request; 401 Unauthorized;

				participant's enrolment.		404 Not Found; 409 Conflict
29	GET	/api/payments/{id}	Retrieves payment details and payment status.	Any	None	200 OK – payment details; 401 Unauthorized; 404 Not Found
30	GET	/api/events/{eventId}/images	Retrieves images associated with an event.	None	None	200 OK – image list; 404 Not Found
31	POST	/api/events/{eventId}/images	Adds an image reference to an event. The image can later be stored in Azure Blob Storage in Part 3.	Organiser	ImageUrl, ImageName	201 Created; 400 Bad Request; 403 Forbidden; 404 Not Found
32	GET	/api/participants/profile/emergency-contacts	Retrieves emergency contacts for the authenticated participant.	Participant	None	200 OK – emergency contact list; 401 Unauthorized
33	POST	/api/participants/profile/emergency-contacts	Adds an emergency contact to the authenticated participant's profile.	Participant	ContactName, Relationship, PhoneNumber	201 Created; 400 Bad Request; 401 Unauthorized

3. API Design Notes

Authentication

RaceDay will use authenticated requests for protected endpoints. A JWT-based approach is planned for Part 2. The authenticated user's identity and role will be used to enforce access control.

Events and Categories

Events are created and managed by Organisers. Each event must have at least one category, such as 5 KM, 10 KM, 21.1 KM or 42.2 KM.

Enrolments

An enrolment connects a ParticipantProfile to a Category. The Category already identifies its Event, so EventID is deliberately not duplicated in the Enrolment entity.

Results

A result belongs to an enrolment. An enrolment may exist before a result is recorded because the participant may have registered before race day.

Routes and Weather

Route and weather endpoints allow participants to prepare for race day. Weather records can later be populated or integrated with an external weather service.

Payments

Payment endpoints represent the event-entry payment process. A future implementation can integrate a real payment provider.

Event Images and Azure Blob Storage

The EventImage entity stores image metadata and a URL. In Part 3, Azure Blob Storage will be used for the actual image files, while the API/database can store the resulting image URL and file name.

4. Consistency With the ERD

The endpoint plan corresponds to the RaceDay ERD entities:

- User
- OrganiserProfile
- ParticipantProfile
- Event
- Category
- Enrolment
- Result
- Route
- Weather

- Payment
- EmergencyContact
- EventImage

The API implementation in Part 2 should follow these routes and responsibilities unless a documented design change is required.